JieZi: A Large-Scale Expert-Audited Dataset and Benchmark for Ancient Chinese Character Exegesis Supplementary Material

This supplementary material provides additional details that could not be included in the main paper due to space constraints. First, Sec. A provides scholarly background on the ACCE task formulation and illustrative examples. Second, Sec. B presents further details on dataset construction, including data distribution, prompt templates, and quantitative verification statistics. Finally, Sec. C provides additional experimental details and the complete results on JieZi-Bench.

A Task Formulation Details

A.1 Scholarly Background of ACCE

Ancient character exegesis. In Chinese paleography, *exegesis* refers to the scholarly practice of interpreting an ancient glyph by jointly analyzing its visual form, internal structure, semantic content, and historical evolution [12, 25, 26]. As one of the oldest continuously used writing systems, ancient Chinese characters carry irreplaceable value for research in archaeology, history, and historical linguistics [8, 11], making their systematic interpretation a long-standing scholarly priority. Unlike modern character recognition, which maps a glyph to a single Unicode label, exegesis requires the scholar to explain *why* a character has its particular form, meaning, and evolutionary trajectory. In practice, scholars follow a progressive analytical workflow [2, 20] that can be decomposed into four sequential stages:

- **Step 1: Identification.** The scholar begins by **identifying the character** and **determining which historical script period** it belongs to.
- **Step 2: Structural Analysis.** The scholar then analyzes the glyph's **internal structure** by **decomposing it into components** and examining how each component contributes to the character's form and meaning.
- **Step 3: Semantic Interpretation.** Building on this structural understanding, the scholar interprets the character's **original semantic value** in its historical context.
- **Step 4: Diachronic Tracing.** Finally, the scholar traces how both **form and meaning evolved** across successive script periods, explaining the motivations behind these diachronic changes.

This multi-stage, knowledge-intensive process constitutes the standard methodology taught in Chinese philology programs and practiced in archaeological and linguistic research.

Challenges in exegesis. Exegesis is particularly challenging because it demands the simultaneous integration of heterogeneous knowledge types. These include visual pattern recognition for identifying glyph components, linguistic knowledge for understanding component functions, semantic reasoning for inferring original meaning, and historical knowledge for explaining diachronic change. Each requires different cognitive skills and a distinct body of expertise. Moreover, ancient glyphs exhibit high visual variability. The same character may appear markedly different across script periods (e.g., OBI, Bronze, Seal, Clerical), and different characters may share visually similar components, making identification and structural analysis error-prone even for trained scholars. The reasoning involved is also inherently layered. Errors in early stages (e.g., misidentifying a component) cascade into incorrect semantic and evolutionary interpretations. As a result, partial understanding is often insufficient, and the task cannot be reduced to independent classification problems. Training a human expert in character exegesis typically requires years of graduate-level study, and the analysis of a single character may involve cross-referencing multiple authoritative dictionaries and reconciling conflicting scholarly interpretations [6, 19].

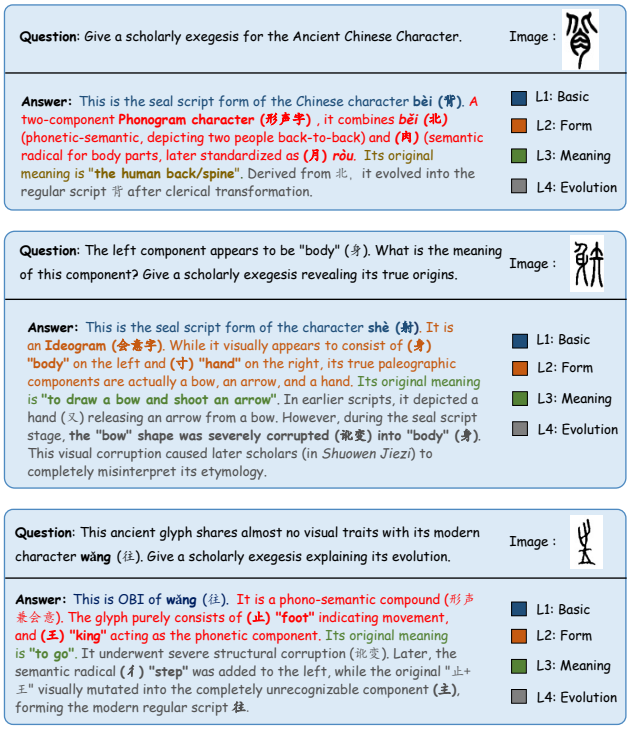

Figure 1: Examples of comprehensive scholarly exegesis in complex paleographic scenarios.

Formalization as a computational task. Despite its importance, exegesis has received limited attention from the computational community. Existing work on ancient Chinese characters focuses predominantly on recognition and retrieval [7, 13], which corresponds to only the most elementary step of the full exegetical workflow. To our knowledge, no prior work has attempted to

COMR

Question: What character is this glyph? Image :

Answer: This is character sī (司). L1: Basic

SCRC

Question: What is the script type of this glyph? (OBI, Bronze, Warring state, Seal, Clerical, Regular, Cursive) Image :

Answer: This is a **Bronze** script. L1: Basic

STRC

Question: What is the structure of this glyph? Image :

Answer: **Left-right structure.** L2: Form

COMR

Question: What is the component on the left? Image :

Answer: 言(yán) L2: Form

COMF

Question: What is the function of the component '贝'? (Semantic, Phonetic, Symbol) Image :

Answer: Semantic L2: Form

COMI

Question: What is the interpretation of the component '贝'? Image :

Answer: It resembles the form of a shell currency, signifying valuables. L2: Form

FORC

Question: Which of the Six Scripts does this glyph belong to?(Pictogram, Ideograms, Ideogrammic compounds, Semantic-phonetic, Derivative, Borrow) Image :

Answer: Pictogram L2: Form

ORIM

Question: What is the original meaning of this glyph? Image :

Answer: It originally refers to **the seasons of the year**. L3: Meaning

COME

Question: What is the type of evolution for the component '厂' in this glyph? (Retention, Corruption, Simplification, Elaboration) Image :

Answer: Corruption L4: Evolution

EVOI

Question: What is the evolution of the character in this glyph? Image :

Answer: Historically, it evolved from an OBI depicting a child in the belly, to a Bronze Script showing a baby gestating in a placenta, and finally to a Seal Script formed by '勹' (wrapping) and '巳' (fetus). The modern standard character presents a semi-enclosed structure, where the outer radical wraps around the inner component. L4: Evolution

Figure 2: Examples of Question-Answering pairs across the ten fine-grained sub-tasks within the ACCE framework.

Question: Give a scholarly exegesis for the Ancient Chinese Character. Image :

Doubao-seed-2.0-pro:
This character consists of three retained semantic components: the '宀'-shaped outer frame representing the granary building, the measuring vessel reflecting its storage function, and the grain (粟) denoting the stored food.

Qwen3.5-9B + JieZi-Dataset
This character is composed of three retained components: the semantic '宀' (representing a house or covering) and semantic '爿' (representing a bed), plus the phonetic '未' which provides the sound.

Question: Explain the historical evolution of the glyph in the image. Image :

Doubao-seed-2.0-pro:
This character is a phono-semantic compound with '心' as the semantic component and '满' as the phonetic component. As it developed into the Regular Script, it retained its top-bottom structure. Its core components have not undergone significant graphical corruption, keeping the overall form regular and stable.

Qwen3.5-9B + JieZi-Dataset
The character was originally written as '慲' (composed of the semantic '心' and phonetic '苒'), with its original meaning being 'vexed' or 'depressed'. A later variant emerged using '满' as the phonetic component. In the modern simplified character, the phonetic part '满' is simplified while the semantic part '心' is retained, finalizing its standard form as '懣'.

Figure 3: Examples of responses for different tasks from different models. Red text indicates error messages, and blue text indicates correct messages.

formalize the complete, multi-stage scholarly process as a computational task. ACCE addresses this gap by decomposing exegesis into four progressive levels that directly mirror the stages of the scholarly workflow described above: identifying a character and its script period (**L1: Basic Information**), analyzing internal structure and component functions (**L2: Glyph Form**), interpreting the original semantic value (**L3: Glyph Meaning**), and tracing diachronic evolution (**L4: Diachronic Evolution**). This decomposition preserves the layered and dependent nature of the original scholarly practice while making each stage independently measurable. It enables researchers to construct targeted supervision, apply stage-specific evaluation metrics, and identify precisely which aspects of exegesis current models can and cannot handle.

A.2 Examples of Tasks

To provide a more intuitive understanding of the ACCE task, we present qualitative examples from the JieZi-Dataset. Fig. 2 illustrates the ten fine-grained subtasks. While these subtasks assess individual capabilities, the ultimate goal of ACCE is holistic interpretation. Fig. 1 presents comprehensive scholarly exegesis in real scenarios, where models are challenged with complex paleographic phenomena such as severe structural corruption and visually similar but

Table 1: Coverage of ACCE subtasks within the JieZi-Dataset. A single comprehensive QA pair may simultaneously address multiple subtasks, the cumulative count across all subtasks exceeds the total number of unique QA pairs (500K).

Level	Subtask	QAs Covering Subtask
L1: Basic	CHAR	175,875
	SCRC	175,875
L2: Form	STRC	175,875
	COMR	175,875
	COMF	221,558
	COMI	221,558
	FORC	312,984
L3: Meaning	ORIM	175,875
L4: Evolution	COME	221,558
	EVOI	210,360
Overall	**Unique QA Pairs**	505,867

semantically distinct glyph forms. As demonstrated, a high-quality exegesis successfully resolves these challenges by synthesizing all four progressive levels (L1 to L4) into a coherent, domain-aligned explanation. Fig. 3 contrasts the responses of the fine-tuned model with those of baseline models on representative subtasks.

B Dataset Construction and Quality Control

B.1 Data Distribution and Metrics

JieZi-Dataset. To illustrate the data diversity and comprehensiveness, Tab. 1 presents the subtask coverage distribution within the JieZi-Dataset. Unlike the individually targeted test queries in JieZi-Bench, the training data often features comprehensive QAs that simultaneously address multiple analytical dimensions, mirroring the holistic nature of real-world scholarly exegesis. Consequently, the sum of subtask frequencies exceeds the total number of unique QA pairs.

JieZi-Bench. Tab. 2 summarizes the question distribution and corresponding evaluation metrics across the ACCE tasks in JieZi-Bench. Notably, the four component-related subtasks (COMR, COMF, COMI, and COME) share the same set of test questions (1,852 per subtask), because each question requires the model to analyze all four component dimensions simultaneously. Within this shared question set, each subtask is evaluated with a distinct metric.

B.2 Prompt for Structurization

Raw dictionary entries are written in dense, unstructured prose that interleaves character identity, structural analysis, semantic explanation, and evolutionary commentary in a single paragraph. Stage 2 (Structurization) aims to distill this unstructured text into compact, structured JSON records aligned with the four ACCE levels (L1–L4), retaining only the information relevant to exegesis while discarding editorial remarks, cross-references, and other content not directly useful for downstream VQA generation.

Table 2: Distribution of Each Task in JieZi-Bench

Task	Metric	Question Count
CHAR	Accuracy	1024
SCRC	Accuracy	1024
STRC	Accuracy	1024
FORC	Accuracy	1024
ORIM	BERTScore	1024
EVOI	LLM-as-a-Judge	1024
COMR	F1-Score + Accuracy	1852
COMF	F1-Score + Accuracy	1852
COMI	F1-Score + BERTScore	1852
COME	F1-Score + Accuracy	1852
Overall	**Unique QA Pairs**	7996

This structurization is non-trivial because ancient Chinese characters exhibit highly heterogeneous complexity. A simple pictograph may require only a brief structural note, whereas a phonosemantic compound spanning multiple script periods may involve dozens of components, variant forms, and evolutionary branches. The prompt must therefore accommodate this wide range of complexity within a single unified schema without losing critical details for complex entries or generating spurious fields for simple ones.

Fig. 4 presents the full prompt used in this stage. Its design addresses three key requirements: (1) *Schema formatting*. The prompt specifies a strict JSON schema with all required field names and value types, ensuring machine-parseable outputs without post-hoc reformatting. (2) *Source fidelity*. The model is explicitly instructed to extract only from the provided text and to leave fields empty rather than fabricate content, which is critical for mitigating hallucination on rare or ambiguous entries. (3) *Adaptive granularity*. The schema naturally accommodates entries of varying complexity: simple characters produce compact records, while complex multi-period entries expand as needed without requiring separate templates.

B.3 Templates for VQA generation

Stage 3 (VQA Generation) transforms the structured metadata from Stage 2 into natural-language VQA pairs suitable for supervised fine-tuning. The generated QA pairs are designed to *simulate the progressive reasoning process of domain experts*, reflecting the layered analytical workflow of scholarly exegesis rather than isolated factual retrieval.

We design four prompt templates corresponding to the four ACCE levels (L1–L4) and adopt a *dual-track generation strategy*. For **factual subtasks** with well-defined answer patterns (e.g., character identity at L1, structure type at L2), QA pairs are produced by directly instantiating the template with the corresponding metadata fields, requiring no LLM reasoning. For **reasoning-intensive subtasks** that must synthesize multiple fields into coherent explanations (e.g., component-level interpretation at L2, semantic analysis at L3, diachronic tracing at L4), LLM generation is necessary. However, general-purpose LLMs lack the specialized paleographic knowledge required for ACCE, and unconstrained generation often produces plausible-sounding but factually incorrect answers.

<Template> Extraction

Role
You are a top-level classical Chinese dictionary data-cleaning expert and structured-data engineer. Your task is to convert noisy, inconsistently formatted semi-structured classical Chinese dictionary text into extremely clean JSON that strictly follows the required specification.

Task
Read the single-character dictionary text from the user. Parse fields such as [Original Character], [Pinyin], [Derived/Variant Characters], [Structure], [Original Meaning], and [Evolution], and output strictly according to the required JSON schema.

1) Global cleanup and noise removal
- Remove meaningless layout symbols, especially stray backslashes mixed into Hanzi (for example, clean `\傍` into `傍`).
- Ignore and drop the entire [Glyph Form] section (from [Glyph Form] to the next section header).
- Ignore all non-schema sections (for example [Illustration], [Character Composition], etc.). Do not mix them into `form_chunk` or `meaning_chunks`.
- Do not output image paths, `[image: ...]`, or anything like `extracted_images/...`.
- Keep example sentences as-is (including parenthetical notes). Only do necessary formatting cleanup.

2) Field mapping and merge rules
- `char`: extract from [Original Character].
- `pinyins`:
  - Extract all pinyin from [Pinyin]. Split multiple values by semicolon/comma/space into an array.
  - Important completion: if [Evolution] contains patterns like "`通 `X(pinyin)' / `又通 `X(pinyin)'", add that pinyin to global `pinyins` (deduplicated).
- `derived_chars`:
  - Mainly from [Derived Characters]/[Variant Characters]. If missing, output `[]`.
  - Split by separators such as `、`, `；`, `，`, `,`, `  `, or spaces.
  - Optional completion: if [Structure] explicitly says "variant written as X / colloquial form X / also written as X", and X is not already listed, add it (deduplicated).
- `form_chunk`:
  - Keep only cleaned original text from [Structure], and keep the `【构造】` tag at the beginning.
  - Do not merge [Original Meaning] text into `form_chunk`.

3) Meaning splitting (most critical)
- [Evolution] contains all senses. Split thoroughly: each numbered sense in the source text (`1.`, `2.`, `3.`...) must become one object in `meaning_chunks`.
- Use an internal state machine:
  - `pinyin` inheritance: when markers like `㊀` / `㊁` with pinyin appear, update current pinyin; following senses inherit it unless a new one appears.
  - If no pinyin marker appears in [Evolution], default all senses to the first item of global `pinyins`.
  - `pos` inheritance: when explicit POS appears (verb/noun/adjective/onomatopoeia etc.), update current POS; following senses inherit it unless changed.
- `meaning` extraction:
  - Remove numbering and leading filler words (like "originally means / refers to / also refers to / describes"). Keep core meaning.
  - If explicit derivation exists, you may append a parenthetical note to meaning.
- `notes` extraction:
  - Put special usage into `notes` (joined by `；`): original meaning, used as "X", interchangeable with "X", old interchangeable form, also interchangeable with "X(pinyin)", also used as surname, etc.
  - Do not move parenthetical text from examples into notes.
- `examples` processing:
  - Content after colon is usually examples. Original examples may be separated by `|`; normalize to `example1 | example2 | example3`.
  - Clean backslashes in examples only; do not alter other wording.
  - If no examples, set `examples` to `""`.
- Non-empty fallback:
  - `pinyin` and `pos` must never be empty. If missing in source, fill `pinyin` with inherited/current (or first global pinyin), and fill `pos` with `other`.
Warning
- Do not miss any numbered sense.
- Do not output any extra characters outside this JSON.

[System lines appended by code in build_messages]
Again: output only one valid JSON object. Do not output code blocks. Do not output explanatory text.
If fields are missing, fill them according to the prompt rules so the schema is complete.

User message
- The user content is `entry_text`, i.e., preprocessed text from each txt file, not a fixed template sentence.
- Preprocessing in `_preprocess_entry`:
  - normalize line breaks and Unicode;
  - remove the full [Glyph Form] section;
  - keep only: original char, pinyin, derived char, variant char, structure, original meaning, evolution;
  - send the cleaned entry text as the user content.

2) Extra system prompt used in retry with larger max_tokens
Your previous output may have been truncated due to length limits, causing incomplete JSON.
Please regenerate the same JSON completely, and it must be strictly parseable JSON.
To reduce length, output compact JSON (no indentation, no extra spaces, no line breaks), but do not omit any field/sense/example.

3) Extra system prompt used in paged fallback
You must use paged output mode.
Output only one valid JSON object, and you may include an extra field `next_index` (integer).
Starting from index {start_index}, output at most {page_size} `meaning_chunks` items in the original numbered order.
Each `meaning_chunks` item must include complete original examples in `examples` (no omission).
If more senses remain, set `next_index` to the next sense index; otherwise set `next_index=0`.
To reduce length, output compact JSON (no indentation, no extra spaces, no line breaks).

Figure 4: Prompt template used for structured metadata extraction from dictionary entries.

Our templates address this by (1) injecting the verified structured metadata as dynamic context, constraining the LLM to reason from the provided evidence rather than its parametric knowledge, and (2) guiding the model through the expert analytical workflow (identification → decomposition → interpretation → evolution tracing). Each template also includes multiple question phrasings per subtask to promote syntactic diversity. Fig. 5 presents a representative template.

B.4 Quantitative Verification Details

To make the quality-control process more transparent, we report the verification scope, sampling strategy, and revision statistics for each stage of our pipeline.

Stage 1: Collection. For JieZi-Dataset, Stage 1 mainly involves digitization, including OCR correction and alignment of the digitized content with the original dictionary pages. Since the goal at this stage is only to ensure fidelity to the source dictionary rather than paleographic interpretation, the full scanned volume (over 2,000 pages) was checked by general annotators without requiring domain expertise. The main error types were OCR omissions and misrecognition of variant character forms. While errors were found in less than 1% of entries, this full manual verification remains indispensable: as noted in the main paper, OCR errors disproportionately concentrate on rare and archaic glyphs, which are precisely the characters most critical to ACCE. Without exhaustive correction, even a small number of such errors would propagate into downstream structuring and QA generation, silently degrading the most informative entries in the dataset.

For JieZi-Bench, the source entries were also fully checked at the collection stage. Errors requiring correction were found in approximately 10% of entries, a notably higher rate than that of JieZi-Dataset. This is expected for two reasons: (1) JieZi-Bench integrates entries from multiple authoritative dictionaries, and cross-source alignment and deduplication introduce additional inconsistencies absent from the single-source Dataset; (2) as a benchmark intended for evaluation, JieZi-Bench applies stricter acceptance criteria for entry completeness and accuracy, causing more entries to be flagged for revision. All identified errors were corrected before proceeding to the next stage.

Stage 2: Structurization. At Stage 2, raw dictionary entries were converted into structured metadata records by an LLM-based extraction pipeline. Given the scale of JieZi-Dataset (over 500K entries), exhaustive manual verification was impractical at this stage. Instead, to verify dataset quality, we randomly sampled 10% of the extracted metadata records. Experts inspected each sampled record by comparing it directly with the source text. Errors were found in approximately 1% of the inspected subset. The most common errors were missing structured fields and cross-source confusion. For example, the pipeline might misattribute a semantic explanation originally associated with the Oracle Bone form to the Bronze Inscription field of the same character.

For JieZi-Bench, all structured metadata records were exhaustively checked by domain experts. Errors requiring correction were found in approximately 5% of entries; all identified errors were corrected before proceeding to VQA generation.

<Template> VQA generation

Please output a structured JSON based on the image. Required fields: glyph type, character formation method, structure, special structure, components, original meaning, diachronic glyph evolution.

Please generate a standard answer that can be used directly for SFT from this single-character image.

Your output must satisfy all requirements below:
1. Output only one JSON object.
2. Use exactly these fields: glyph type, character formation method, structure, special structure, components, original meaning, diachronic glyph evolution.
3. The field "glyph type" must be consistent with the sample label "{glyph type}".
4. "structure" and "components" must follow the actual form in the current image stage.
5. "original meaning" and "diachronic glyph evolution" must be grounded in context, especially historical divergence, later forms, and simplification merges.
6. For simplified-form images, modern mainstream meaning is allowed for "original meaning", but the source chain must be explicit in "diachronic glyph evolution".
7. If weak hints conflict with the image, trust the image.
8. "original meaning" must align with the candidate list first; only minor paraphrase is allowed when strongly supported.
9. If "structure" is "single component", "special structure" must be a short description; otherwise it must be an empty string.
10. If the same component appears multiple times in different positions, split it into separate keys with position suffixes.

Recommended internal order:
- Identify the glyph stage.
- Identify the actual structure in this stage.
- Identify visible components and their functions.
- Determine the original meaning.
- Complete the diachronic evolution chain.

Current sample info:
- Sample ID: {sample_id}
- Image filename: {image_name}
- Weak glyph label hint: {glyph_type}

Failure correction hint:
{retry_guidance}

Auxiliary context (for reference, not for literal copying):
{labeling_context_json}

Please identify the modern dictionary headword shown in the image.

What is the dictionary headword of the character in this image?

Please answer the headword of this image directly.

Please classify the glyph type of the character in the image (e.g., Oracle Bone Script, Bronze Script, Seal Script, Clerical Script, Regular Script, Warring States Script, Simplified).

Which glyph type does this image belong to?

Please provide the glyph category of the character image.

What is the modern dictionary headword corresponding to this glyph image?

Please identify the modern dictionary headword of the character in the image.

Based on the image, what is the character headword in a modern dictionary?

Which glyph type is shown in the image?

...

Figure 5: Prompt template for VQA generation. The prompt integrates to generate high-quality SFT data.

Table 3: Training settings and hyper-parameters for different model scales.

Setting	Qwen3.5-2B	Qwen3.5-4B	Qwen3.5-9B
Batch size	128	128	64
Learning rate	1e-5	1e-5	1e-5
LR schedule	cosine	cosine	cosine
Warmup ratio	0.05	0.05	0.05
Weight decay	0.1	0.1	0.1
Optimizer	AdamW	AdamW	AdamW
Max length	2048	2048	1024
Epochs	5	5	5
Precision	bf16	bf16	bf16
Hardware	8 910B	8 910B	8 910B
Training time	96 NPU hours	192 NPU hours	384 NPU hours

Stage 3: VQA Generation and Verification. At Stage 3, VQA pairs were generated from the verified metadata by an LLM guided by expert-designed templates. The use of templates is essential because unconstrained LLM generation would suffer from three problems: (1) *format inconsistency*, where question phrasing and answer granularity vary unpredictably, hindering standardized evaluation; (2) *uneven subtask coverage*, where the LLM tends to favor subtasks it handles well while underrepresenting harder ones such as COME and EVOI; (3) *hallucination risk*, where the LLM may fabricate etymological explanations or evolutionary paths absent from the source dictionaries. Our template-guided approach addresses all three issues. Each template prescribes the question format and expected answer scope for a given subtask, ensuring uniform coverage and consistent granularity. The LLM then generates natural-language VQA pairs grounded in the metadata fields specified by the template, combining the scalability of LLM generation with the controllability of structured constraints.

Despite these safeguards, template-guided generation introduces its own errors. The generation process can produce image-content mismatches (e.g., pairing a glyph image with an answer describing a different script period), and natural-language answers may deviate from the source text in phrasing or scope. Moreover, VQA-level verification covers broader dimensions than Stage 2, including not only field-level correctness but also the naturalness of question-answer formulations and the appropriateness of paired images, which increases the likelihood of flagging issues.

For JieZi-Dataset, we randomly inspected 5,000 VQA instances. Since the underlying metadata had already been verified at Stage 2, the sampling ratio was reduced accordingly. Errors were found in approximately 3% of the inspected subset; the slightly higher rate compared to Stage 2 reflects the new error sources and broader verification scope described above rather than upstream propagation.

For JieZi-Bench, all VQA pairs were manually checked, since benchmark reliability is critical for evaluation. Errors requiring correction were found in approximately 10% of entries, consistent with the pattern observed at earlier stages: the benchmark's stricter acceptance criteria and cross-source complexity lead to a higher correction rate.

Table 4: Ablation study on different data construction stages. Struct.: structured metadata; VQA: After VQA generation; Open-src: VQA with open-source data augmentation.

Struct.	VQA	Open-src	CHAR ↑	COMF ↑	COMI ↑	EVOI	
						FAC ↑	SCE ↑
			18.9	13.7	17.0	7.9	20.2
✓			27.0	29.7	24.1	15.4	26.3
✓	✓		35.3	31.4	37.1	25.6	31.9
✓	✓	✓	41.8	32.7	41.3	27.9	36.7

Overall, JieZi-Dataset adopts random spot-checking at Stages 2 and 3 to balance scale and quality, while JieZi-Bench uses exhaustive manual verification at all stages. These combined strategies ensure that JieZi-Bench achieves expert-level reliability for evaluation, while JieZi-Dataset maintains sufficient quality for training at scale.

C Experimental Details and Full Results

C.1 Additional Training Details

Table 3 summarizes the hyperparameter settings for instruction tuning. All experiments were conducted on 8 Ascend 910B NPUs.

C.2 Ablation Studies

Tab. 4 reports the impact of each data construction stage on Qwen3.5-2B. The first row corresponds to the few-shot base model without fine-tuning. We select CHAR, COMF, COMI, and EVOI as representative subtasks, as they span all four ACCE levels and cover both classification and generation objectives.

Each stage contributes a distinct gain profile. Structured metadata produces the largest improvement on the classification-oriented COMF (+16.0), since explicit component-function labels offer dense supervision that directly matches this subtask. In contrast, gains on generative subtasks remain moderate (COMI +7.1, FAC +7.5). VQA reformatting reverses this pattern: the generative subtasks benefit most (COMI +13.0, FAC +10.2), whereas COMF improves by only +1.7. This contrast suggests that recasting structured records as natural-language QA pairs trains the model to *articulate* multi-step reasoning rather than simply retrieve labels.

Open-source augmentation, in turn, primarily strengthens visual robustness (CHAR +6.5) and scholarly expression (SCE +4.8). Factual subtasks show smaller gains (COMF +1.3, FAC +2.3), confirming that their performance is bottlenecked by knowledge rather than visual diversity. Taken together, the three stages address complementary dimensions of factual grounding, reasoning articulation, and visual robustness. Their consistent, non-overlapping improvements validate the necessity of each stage in the proposed pipeline.

C.3 Human Validation of the LLM-as-a-Judge

To rigorously validate the reliability of our LLM-as-a-Judge protocol for the Evolution Interpretation (EVOI) task, we conducted a human-LLM agreement study. We randomly sampled 200 responses in total across 4 representative models to ensure a diverse distribution of response qualities. Three human experts with backgrounds in Chinese paleography independently evaluated these responses on

Table 5: Correlation between human expert ratings and LLM judge scores on the EVOI task.

Metric	Pearson (r)	Spearman (ρ)
FAC	0.68	0.71
SCE	0.61	0.64

a scale of 1 to 5, focusing on two dimensions: Fact Alignment (FAC) and Scholarly Expression (SCE).

We computed the Pearson (r) and Spearman (ρ) correlation coefficients between the automated LLM judge scores and the human ground truth. As shown in Tab. 5, the LLM judge exhibits strong correlations (> 0.60) with expert evaluations across both metrics. The results confirm that the LLM judge aligns closely with human scholarly standards, demonstrating its effectiveness and reliability as an automated evaluation metric for ancient character exegesis.

Judging prompts. We provide the full prompts used in our LLM-based judging protocol for EVOI. Fig. 6 presents the prompt for evaluating **Fact Alignment**, while Fig. 7 presents the prompt for evaluating **Scholarly Expression**. We used Doubao-seed-2-0-lite-260215 [4] as our LLM-as-a-judge model.

C.4 Main Results on JieZi-Bench and Analysis

Due to space limitations in the main text, we present the complete experimental results in Table 6, including additional models not shown in the main paper. Below, we provide a detailed analysis of the key findings.

Scaling law holds within model families. The supplementary table includes additional model variants not reported in the main paper, enabling within-family comparisons among general MLLMs. Across all families, larger variants consistently outperform smaller ones. Gemini 3.1 Pro surpasses Gemini 3.1 Flash on every metric (e.g., CHAR: 29.9 vs. 25.7; SCRC: 76.6 vs. 56.5), Qwen3.5-397b-a17b outperforms 35b-a3b (e.g., SCRC: 67.1 vs. 61.4), and InternVL3.5-8B consistently exceeds its 4B counterpart (e.g., ORIM: 58.0 vs. 24.3). These within-family comparisons, together with the cross-family results in the main paper, confirm that the general scaling trend holds for ACCE.

Domain-specialized pretraining does not guarantee ACCE proficiency. tonggu-vl-2B [5] is a model specifically fine-tuned on massive ancient Chinese document corpora for tasks such as text recognition and reading comprehension. Despite this domain-specialized pretraining, it scores below 5% on most ACCE subtasks (e.g., CHAR: 3.1, STRC: 0.2, COMR/COMF/COMI: 0.0) and achieves only 12.9 on ORIM. This reveals a critical distinction: *recognizing and reading ancient texts* is fundamentally different from *performing scholarly exegesis*, which requires structured reasoning about glyph form, component function, semantic origin, and diachronic evolution. The failure of tonggu-vl demonstrates that ACCE poses genuinely novel challenges beyond the reach of existing domain-specific models, thereby validating both the task formulation and the necessity of JieZi-Dataset as a dedicated training resource. Conversely, once equipped with JieZi-Dataset, even a 9B model outperforms general MLLMs that are one to two orders of magnitude larger (e.g., Kimi-K2.5 170B, GLM-4.6V 107B, Qwen3.5-397b) on nearly every subtask, further confirming that high-quality domain-specific data is a far more effective lever than model scale alone for knowledge-intensive tasks like ACCE.

Remark on the anomalous behavior of Qwen3.5-4B. As shown in Tab. 6, few-shot Qwen3.5-4B scores near zero across all

<Template> LLM-judge FAC

You are a strict evaluator for ancient Chinese character tasks.
This time, you will evaluate only one dimension: fact alignment.
Your task is not to evaluate the entire response. You must only evaluate whether the sentences describing the "historical character shape evolution" in the prediction are factually consistent with the "historical character shape evolution" in the reference answer.
You must adhere to the following principles:
Base your judgments solely on the provided Ground Truth. The use of external knowledge is strictly prohibited.
Do not evaluate writing style, wording, or eloquence.
Do not award high scores simply because the prediction "sounds like" a professional ancient character analysis.
Differences in phrasing from the reference answer are acceptable, but altering factual information is not.
ground_truth_full serves only as auxiliary background context. The primary targets for scoring
remain ground_truth_evolution and prediction_evolution.
Partial credit is allowed for this dimension: if the prediction has substantial points of alignment regarding key structures, component relationships, historical periods, or evolutionary directions, award corresponding points. Do not automatically penalize the score to 0 or 1 merely because some errors exist.
Please focus your review on the following:
Alignment of the core structural composition (核心构形).
Alignment of relationships between key components (关键部件关系).
Alignment of key historical periods/stages (关键时代阶段).
Alignment of the evolutionary direction (演变方向).
Whether the character has been conflated with a different character or an entirely different structural system.

Scoring Rubric (Must be exactly 0, 1, 2, 3, or 4):
4: Highly consistent with the reference answer; almost zero factual errors.
3: The main narrative is largely consistent, but there are some factual deviations or inaccuracies in minor details.
2: Contains verifiable and clear points of alignment (e.g., correctly identifies at least *some* structures/components/periods/directions), but concurrently contains obvious factual errors. Overall reliability is moderate.
1: Shows only minimal, superficial similarity; the vast majority of key facts are incorrect.
0: Fundamentally contradicts the reference, is completely blank, or describes the shape evolution of an entirely different character/system.

Mandatory Calibration Rules:
If the prediction is blank: score = 0
If the core structural composition, core components, or main evolutionary direction are vastly incorrect: generally score <= 2
If the character is misidentified as a different structural system, or the main factual narrative is fundamentally misplaced: score = 0
If the response relies purely on generic clichés/filler with almost no verifiable facts: the score cannot exceed 2
If the prediction contains multiple key points that align with the reference (even if accompanied by errors), the score generally should not be penalized to 0 or 1.
Do not default to grading most samples as 3 or 4 (avoid score inflation).

Now, please evaluate the following sample to be rated according to the scoring rules above.
Ground Truth: {}
Output of model: {}

Figure 6: Prompt template for evaluating fact alignment within our LLM-as-a-judge protocol.

Table 6: Main results on JieZi-Bench across four paleographic levels: L1 (Basic Info: CHAR, SCRC), L2 (Glyph Form: STRC, COMR, COMF, COMI, FORC), L3 (Meaning: ORIM), and L4 (Evolution: COME, EVOI-FAC/SCE). All metrics are scaled to 0-100.

Method	L1		L2					L3	L4		
										EVOI	
	CHAR↑	SCRC↑	STRC↑	COMR↑	COMF↑	COMI↑	FORC↑	ORIM↑	COME↑	FAC↑	SCE↑
Closed-source MLLMs											
GPT-5.4 Thinking [17]	10.4	39.2	50.4	13.8	11.9	10.3	33.1	59.9	12.4	10.2	16.8
Gemini 3.1 Pro [10]	29.9	76.6	65.7	31.9	29.9	24.8	56.1	62.8	30.3	22.3	27.1
Gemini 3.1 Flash [9]	25.7	56.5	53.5	24.6	22.5	19.1	44.8	54.1	23.2	17.5	23.1
Claude opus 4.6 [1]	22.9	65.8	63.5	28.0	26.3	22.3	50.6	59.3	26.7	19.8	26.3
Doubao-Seed-2.0-pro [3]	36.6	68.5	69.8	37.8	34.4	29.9	59.9	63.2	36.1	31.7	38.7
Open-source MLLMs											
Kimi-K2.5 [21] (170B)	31.5	70.1	68.8	39.2	36.6	31.5	58.1	66.6	36.4	28.3	34.6
GLM-4.6V [22] (107B)	18.2	49.7	48.9	21.8	20.4	18.2	42.2	48.2	20.9	15.5	20.2
Qwen3.5-397b-a17b [18]	26.2	67.1	69.4	32.1	29.9	26.5	54.4	59.5	28.3	22.2	30.7
Qwen3.5-35b-a3b [18]	26.6	61.4	66.5	32.4	29.3	27.3	57.8	65.2	31.0	19.6	25.6
InternVL3.5-8B [23]	00.1	09.0	14.6	0.0	0.0	0.0	21.4	58.0	0.0	24.8	0.1
InternVL3.5-4B [23]	0.0	0.0	0.9	0.0	0.0	0.0	8.2	24.3	0.0	0.0	0.0
tonggu-vl-2B [5]	3.1	05.9	0.2	0.0	0.0	0.0	4.6	12.9	0.0	1.1	0.2
Qwen3.5-2B [18]	18.9	22.6	31.6	17.3	13.7	17.0	30.7	62.3	12.9	7.9	20.2
Qwen3.5-2B + JieZi-Dataset	41.8+22.9	61.7+39.1	72.7+41.1	42.6+25.3	32.7+19.0	41.3+24.3	63.3+32.6	65.7+3.4	28.3+15.4	27.9+20.0	36.7+16.5
Qwen3.5-4B [18]	3.6	6.6	1.0	4.5	3.9	4.3	5.9	6.5	3.6	2.0	3.6
Qwen3.5-4B + JieZi-Dataset	49.0+45.4	67.6+61.0	74.1+73.1	47.2+42.7	37.9+34.0	42.4+38.1	64.8+58.9	65.9+59.4	33.2+29.6	28.5+26.5	37.2+33.6
Qwen3.5-9B [18]	22.3	51.0	40.2	27.4	24.6	26.1	49.1	64.7	22.4	15.0	23.5
Qwen3.5-9B + JieZi-Dataset	**52.6**+30.3	**80.1**+29.1	**74.7**+34.5	**48.1**+20.7	**38.4**+13.8	**45.7**+19.6	**66.5**+17.4	**66.7**+2.0	**45.4**+23.0	**32.5**+17.5	**40.6**+17.1

eleven ACCE subtasks (e.g., CHAR: 3.6, STRC: 1.0, ORIM: 6.5), performing substantially below both the smaller 2B and the larger 9B variants. We verified that all three models share identical inference pipelines, prompts, and decoding parameters, ruling out implementation errors. Manual inspection of the 4B outputs (Figure 8) reveals that the model tends to reproduce content from the few-shot demonstrations rather than generating answers grounded in the current query, producing outputs that bear no meaningful correspondence to the target glyph.

Since Qwen3.5's training details are not publicly disclosed, the root cause remains unclear. We identify two plausible contributing factors. (1) *Copying bias in few-shot ICL*: the model reproduces demonstration content instead of reasoning about the current query. This behavior directly aligns with prior findings that LLMs can develop a copying bias under in-context learning, reproducing surface patterns from examples rather than inducing the underlying task [15, 16]. (2) *U-shaped scaling*: the 2B (functional) → 4B (collapsed) → 9B (functional) pattern is consistent with the U-shaped scaling phenomenon, where medium-sized models are drawn toward easier competing behaviors instead of the target task [14, 24].

After fine-tuning on JieZi-Dataset, Qwen3.5-4B fully recovers to the expected scaling order, achieving the largest absolute improvements among the three model scales (e.g., STRC: +73.1, ORIM: +59.4 vs. 2B's +41.1/+3.4 and 9B's +34.5/+2.0). This confirms that the bottleneck is not model capacity but misaligned few-shot behavior, which domain-specific supervision effectively overrides [27].

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<Template> LLM-judge SCE

You are a strict evaluator for paleography tasks, evaluating only the scholarly expression dimension.
Evaluate **only** the "historical character shape evolution" sentences in the prediction. Judge if the expression is professional, rigorous, and matches a paleographic dictionary (lexicographical) style, rather than reading like layman's explanations or empty clichés.
Core Principles:
GT Only: Base judgments strictly on the provided Ground Truth. No external knowledge.
Style over Fact (Mostly): Evaluate terminology, logic, and professional tone. However, professional tone wrapped around a completely wrong evolutionary narrative cannot score high.
Quality > Quantity: Do not reward length unless it provides valid information gain.
Dictionary Style: Prefer concise, objective summaries of structure and evolution. Penalize lyrical, heuristic, or layman-oriented tones.
Scoring Rubric (Strictly 0, 1, 2, 3, or 4):
4: Highly mature, dictionary-level expression. Precise terminology, restrained judgments, rigorous logic, and substantial content with zero fluff.
3: Professional and solid. Clear lexicographical traits and good logic, but contains minor looseness or slight layman phrasing.
2: Readable but lacks true professional depth. Unstable style, leans toward generic explanatory text, or contains obvious verbosity/clichés.
1: Loose and verbose. Strong layman tone, high fluff, lacks rigor, or merely dresses up generic content with scattered jargon.
0: Blank, or entirely empty clichés/slogans with no valid professional information.
Mandatory Calibration Rules:
If blank: score = 0
If longer than GT but mostly adds repetitive fluff/no new info: score <= 2
If using generic layman phrases (e.g., it can be seen, shows the wisdom of the ancients): score <= 2
If built on obviously wrong factual narratives or incorrect structural judgments: score <= 2
If terminology is chaotic or reads like a generic essay: score <= 1
If merely using jargon like clericalization to mask empty/wrong content: score <= 1
Be Strict: 4 is for exceptionally mature, dictionary-quality text only. Do not default to 3. Do not give high scores just because the text is long or reads smoothly.

Now, please evaluate the following sample to be rated according to the scoring rules above.

Ground Truth: {}

Output of model: {}

Figure 7: Prompt template for evaluating scholarly expression within our LLM-as-a-judge protocol.

Prompt:

Role
Expert in Chinese paleography, Shuowen Jiezi and etymology…

Task

Output a strict JSON component analysis of the ancient character in the image, detailing each component's function, explanation, and evolution type …

Rules:

1. Rely SOLELY on the image; no external knowledge.
2. Never hallucinate unsupported details.
3. Output exactly ONE raw JSON object (NO markdown, NO comments, NO extra text).
4. Prioritize valid fields, aligned content, and conservative analysis. …

Required JSON Format
构件: Object of 1-4 key components.
Keys: Single characters/radicals (use brief desc ONLY if unnameable).
Values: Objects containing exactly:
功能: String or Array of: 表意, 表音, 符号.
演变类型: Choose ONE: 保留, 讹变, 简化, 区别, 繁化.
解释: Brief explanation …

Few-shot1: …
Few-shot2:

```
"构件": {
    "构件名": {
      "功能": "表意/表音/符号",
      "演变类型": "保留/讹变/简化/区别/繁化",
      "解释": "简短解释"
      }
  }
```

Qwen3.5-4B:

Just Copy !

```
"构件": {
    "构件名": {
      "功能": "表意/表音/符号",
      "演变类型": "保留/讹变/简化/区别/繁化",
      "解释": "简短解释"
      }
  }
```

Figure 8: An example of anomalous output from few-shot Qwen3.5-4B, illustrating the copying bias.

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